

15.8 Answers

① a) $\frac{64\pi}{3}$

b) 7π

c) $\frac{9\pi}{4} (2 - \sqrt{3})$

② a) $\frac{2}{35}$

b) $\frac{65\pi}{4}$

c) $\frac{2\pi}{5}$

③ a) $\frac{4\pi}{5}$

b) $\frac{15\pi}{16}$

c) 0

④ $\frac{8\sqrt{2}\pi}{3}$

⑤ $\frac{1}{9}\pi a^3$

⑥ $\frac{1}{96}$

⑦ $\frac{243}{5}\pi$

⑧ hint: $\int_0^{2\pi} \int_0^{\pi/6} \int_0^4 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$

is the set up.

⑨ $\frac{4}{3}\pi a^3$